Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/09/2012
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Long-Term Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ajwyrobek@lbl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	925-989-4466
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Rd MS: 74R0157		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
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Start Date:	01/01/2011	**End Date:**	12/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Loftus, David	**Contact Phone:**	650-604-1011
Contact Email:	david.j.loftus@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	The key personnel on the project are Andrew J. Wyrobek (PI), Trent Northen (metabolomics), Ewa Witkowska (proteomics), Antoine Snijders (cell signaling, bioinformatics), Sandhya Bhatnagar (tissue biology). Dr. Lowe terminated at the Lab in 2012 to return to her medical practice.		
COI Name (Institution):	Snijders, Antoine (Lawrence Berkeley National Laboratory) Northen, Trent (Lawrence Berkeley National Laboratory) Witkowska, Ewa (University of California, San Francisco) Bhatnagar, Sandhya (Lawrence Berkeley National Laboratory) Straume, Tore (NASA, Ames) Rabin, Bernard (UMBC) Britten, Richard (UVMS)		
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Task Description:	The objectives of this research are to characterize the persistent defects in neuro-signaling and cellular damage to central nervous system (CNS) tissue after cranial exposure to space radiation and to identify the critical molecular changes associated with individual variation in persistent neurocognitive deficits, using the rat model. We reported global changes in transcript expression profiles in the hippocampus and choroid plexus (CP) tissues of old animals that were exposed to 56Fe or 12C beams in their youth. We identified changes in the predicted protein expression in the CP that correlated with the magnitude of persistent neurocognitive deficit (i.e., to ~22 months of age). We are now conducting comparative analyses of radiation-induced changes in CP signaling and cellular architecture that are associated with neurocognitive deficits, using CNS tissue collected from animals exposed at various doses of 56Fe or 12C after they were tested for neurocognitive deficits. We are also investigating the early onset of radiation-induced changes to hippocampal and CP signaling, CP architecture, hippocampal neurogenesis, and composition of the cerebrospinal fluid (CSF) at ~3 months after exposure to both high and low-fluence 56Fe, and examining the association between molecular defects and neurocognitive changes within each irradiated animal. Our hypothesis is that radiation-induced damage to the CP can persist for the life of the individual, producing abnormal CSF that lacks critical metabolites and neurogenic factors needed for a normal CNS microenvironment, neurogenesis and cognitive functions. The Specific Aims are: Aim 1: Characterize the dose dependence of persistent signaling and cellular defects in the CP and hippocampal zones of neurogenesis of 12C- and 56Fe-exposed rats that have persistent deficits in their cognitive function and motor behavior. Aim 2: Characterize the time course of cellular and signaling defects in the CP of rats irradiated with 56Fe beams and determine the temporal relationship between CP defects and abnormal hippocampal neurogenesis. Aim 3: Determine whether irradiated animals with defective CP also produce CSF deficient in critical metabolites or proteins that are critical for adult neurogenesis, using proteomic, metabolomic, and neurosphere cultures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research lays a new foundation for understanding the effects of space radiation on the choroid plexus (CP), a small organelle in the CNS ventricles that produces cerebral spinal fluid to nourish the CNS microenvironment. Our project will characterize the time and dose dependence of radiation-induced changes in CP signaling and architecture and how these changes are associated with neurocognitive dysfunctions in individuals exposed to space radiation. Our research will also enhance our understanding of molecular changes in the CP in normal aging and its association with age-related defects in neurocognition.
	2012 TASK PROGRESS AIM 1: Characterize the dose dependence of persistent signaling and cellular defects in the CP and hippocampal zones of neurogenesis of 12C- and 56Fe-exposed rats that have persistent deficits in their cognitive function and motor behavior. 1.1 Tissue sources used in this project. This project utilizes blood, cortex, hippocampal, cerebral spinal fluid (CSF), and choroid plexus (CP) tissue from rats at various times after exposure to various doses of 12C or 56Fe beams. The animal exposures were performed at Brookhaven National Laboratory, NASA Space Radiation Laboratory (BNL-NSRL). The majority of the rats used in this study were also tested for the effects of space radiation on neurocognitive functions. 1.2 Transcript profiling of the rats exposed to HZE space radiation and tissue validation. We found persistent neurocognitive deficits in old rats that had been exposed to HZE radiation (56Fe and 12C) at young ages and hypothesize that the persistent CNS deficits are due, in part, to residual radiation damage of the CNS microenvironment. The old animals showed persistent damage to the vessel wall architecture of the choroid plexus (CP), especially after 1Gy 56Fe exposures ($p < 0.01$). Expression profiling of the cortex and hippocampus of the 56Fe exposed rats showed a significant Huntington's disease subnetwork and identified a three-genes signature that persisted to ~18 months after 56Fe exposed animals, with levels of expression in tissue that correlated with degree of cognitive impairment (R2 ~0.8). The expression of transthyretin (Ttr), which prevents Ab amyloid fibril formation in CNS, showed down-regulation at ~ 1.5 years post exposures ($p < 0.001$) in CP of 56Fe and 12C irradiated rats. These findings identify the CP, which expresses Ttr and produces CSF as a CNS target for persistent radiation damage that may lead to poor memory and behavior in old age. 1.3. Choroid plexus (CP) molecular histopathology. We found histological changes in both the epithelium and endothelium of the CP associated with normal aging and endothelial changes at ~20 months after cranial radiation with 56Fe beams. The vessels of the irradiated animals showed a thickening of the endothelial vessel walls of two types: (1) those with acellular matrix between the RBCs and endothelial lining, and (2) those with additional cellular stroma between endothelium and basement membrane. We are now addressing the question: what are the molecular components of the thickened endothelial vessel walls of the CP of irradiated rats and how are they related to time after exposure, dose of exposure and neurocognitive test results from the same animal? For the remaining two months of 2012, and for the third year of the project we plan to extend the microarray expression profiling and CP expression analyses to tissue from rats treated with 12C. We will continue the studies of the molecular defects of CP endothelial and epithelial linings of irradiated animals and determine whether these defects are seen at both 3 and 18 months after exposure and after low-fluence exposures. AIM 2. Characterize the early response and time course of cellular and signaling defects in the CP of rats irradiated with 56Fe and determine the temporal relationship between CP defects and abnormal hippocampal neurogenesis. In July 2011, we performed a large rat exposure experiment at BNL, exposing middle-aged rats (i.e., retired breeders, ~ 6 months of age) to whole body 56Fe beam at two fluences (10 and 100 cGy plus sham exposed animals; ~15 males per group; plus ~10 young adult males as reference). Animals were then housed at Berkeley Labs for 3 months until October 2011. Each animal was sampled for CSF at 2 months post-exposure (a live surgical procedure) and at 3 months post exposure, just before euthanasia. At 3 months post exposure, we also collected cardiac blood plasma, hippocampus

Task Progress:

and cortex tissue, and one hemisphere was paraffin-embedded. The CSF collected was used for the proteomic and metabolomics studies of Aim 3. These studies are in progress. In October 2012 (last month), we conducted a second large rat exposure study at BNL (sham, 10cGy, 100cGy of 56Fe at 1000Mev/n), to address the following research questions:

• What are the behavioral deficits at 1-2 months after exposure and what is the association between the two behavioral endpoints in the same animals?

• What are the CNS molecular and cellular effect of radiation exposure on hippocampus and CP at 3-4 months after exposure, and how do the molecular and tissue effects relate to the behavioral deficits in each animal?

• What are the proteomics and metabolomics changes in blood plasma at 24 hours and at 3-4 months after radiation, and are the changes in blood markers associated with behavioral, hippocampal and CP endpoints within the same animals?

• What is the relationship between the effects of radiation and changes with normal aging for behavioral, blood and CNS endpoints?

AIM 3. Determine whether irradiated animals with defective CP also produce cerebrospinal fluid (CSF) deficient in critical metabolites or proteins that are important for adult neurogenesis, using proteomic and metabolomic technologies and neurosphere cultures. The starting effort for this aim was to identify significant and biologically plausible changes using metabolomics and proteomic profiles in CSF of the 56Fe irradiated animals collected at 2 or 3 months after exposure.

3.1. Metabolomics of CSF. We investigated CSF samples collected from unirradiated animals for method optimization and for assessing baseline animal-to-animal variation. We report our findings for the 49 CSF samples derived from 39 animals from the four experimental groups (rats exposed to 10 and 100 cGy 56Fe, sham exposed and age matched, and young untreated animals). The Berkeley NASA metabolomics program has focused on gas chromatography coupled to time-of-flight mass spectrometry (GC-TOF) and also included liquid chromatography coupled to quadrupole time-of-flight mass spectrometry (LC-QTOF). One of the central challenges of this research program was the limited volume of CSF available from animal studies. Initial experiments were aimed at determining the minimum volume that gave good metabolite coverage and reproducibility. Here we used GC-TOF to characterize different volumes of aliquots of CSF 15-100uL and we were pleased to find that 30uL was sufficient to detect a wide range of metabolites with an average CV of <20%. Our metabolomics analyses suggested that 3-hydroxypyridine may be significantly elevated in the 100cGy vs. Sham (t-test p=0.0125), but this difference was not significant when correcting by multiple-hypothesis testing (fdr = 0.8145). The lack of significant changes in metabolomics targets after exposure to ionizing radiation may be explained by insufficient signal to noise and low statistical power, due to the variation among the animals in relation to effects that might be attributable to radiation. Thus, given the numbers of animals in our study and their biological variation, we cannot exclude the possibility that radiation exposure did induce changes of <70% in the level of CSF metabolites.

Therefore, we are shifting our effort to focus on the effects of space radiation on CNS tissue using frozen hippocampal tissue from irradiated rats tested for neurocognitive deficient and as reference, also analyzing blood samples collected before and 24-hour after exposure. This is a result of our collective positive experiences working with tissues coupled with the likelihood that (a) metabolic differences will be more extreme in these tissues and that (b) any tissue finding may be used to construct more targeted assays for the blood and CSF fluids. Long-term plans will focus on performing metabolite imaging experiments to examine the distribution of altered metabolites within the CNS tissues, specifically the areas of the hippocampus and CP. Nanostructure-initiator mass spectrometery will be applied to frozen brain sections using the methods we recently described.

3.2. Proteomics of CSF. Our mass spectrometry quantitation analysis employed iTRAQ™ reagents that allow for concomitant analysis of up to 8 samples. The relative quantitation is based on reporter ions generated from the iTRAQ tags in the process of tandem MS and protein identification is achieved by matching sequence-specific fragment ions to the protein database. To enhance the depth of proteome coverage, an extensive sample fractionation was performed using a 2D LC strategy of combining alkaline and acidic reversed phase chromatography, generating 10 fractions from each iTRAQ multiplex for LC MS analysis. The 48 rat CSF samples were grouped into 3 and 4 multiplexes and the relative protein ratios are based on iTRAQ reporter ion intensities that were normalized to the level of albumin in each sample.

A total of 932 entries were identified above the local false discovery rate (FDR) of 5% using ProteinPilot™ search engine. After consolidating orthologous entries, 402 distinct mammalian proteins were identified. We investigated the effects of radiation on protein concentration in CSF using three comparison groups: 100 cGy vs sham, 10 cGy vs sham, and 100 cGy vs 10 cGy. Pairwise comparisons between the experimental groups were performed using t-Test showed that the majority of differences in relative abundances were modest. It is important to understand, however, that quantitation methods based on iTRAQ chemistry tend to flatten protein ratios and hence some observed changes might be artificially low and hence should not be ignored, especially when they were derived from large group of animals. We identified ~30 CSF proteins per comparison group with significant concentration differences associated with radiation exposure ($p<0.05$). We also investigated the effects of age by comparing the protein concentrations in young animals versus those in sham animals. The latter were age matched to the animals that were irradiated. We identified 48 proteins with significant differences between the older and young animals ($p<0.05$). Bioinformatic annotation of the biological functions of radiation effects and aging are in progress. The top radiation network was associated with “Neurological Disease, Psychological Disorders, Cell Death and Survival” and the top networks for an age effect were: “Cardiovascular Disease, Cell-To-Cell Signaling and Interaction, Hematological System Development and Function” and “Cellular Movement, Developmental Disorder, Neurological Disease.” Further bioinformatics analyses of our proteomic findings are in progress.

Our bottom-up shotgun strategy for discovery proteomics was geared toward detecting global, aggregate changes in CSF protein expression and/or degradation rates and it is considered an important first step in characterizing a biological system. The analysis of rat CSF was successful technically by providing information on over 400 proteins in CSF from a small volume of material. We detected several biologically interesting proteins and functions that are worth pursuing further. However, we did not detect any potent CSF biomarkers of exposure to ionizing radiation with high level of confidence, even though significant tissue expression changes were observed in the CP in the same animals. This suggests that CSF may not be the optimal tissue to discover radiation responsive-protein markers. Therefore, we propose

	to shift the emphasis of our proteomic analysis from CSF to hippocampal tissues and peripheral blood. These studies will be coordinated with the metabolomics analyses to utilize the same experimental animals and exposure groups, whenever possible. Our analytical focus is now on in-depth bioinformatics analyses and in situ validation of the existing CSF proteomic data. We will also prepare for the analysis of tissue that will be available early next year from the CNS and blood from irradiated rats. Once putative biomarkers are identified and validated, both CSF and/or blood could be examined by a specialized analysis of molecules of interest with a goal of developing clinically suitable assays, using either mass spectrometry or immunohistochemistry approaches. In addition, radiation might also produce subtle rather than substantial alterations in proteome, e.g., changes in the repertoire of posttranslational modifications and/or proteolytic processing. To detect these structural changes, alternative targeted approaches are necessary and should be considered in future studies.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
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Abstracts for Journals and Proceedings	Wyrobek AJ. "Choroid plexus, neurogenesis, and neurocognitive performance after exposure to space radiation." Presented at the 2012 NASA CNS Workshop, January 17, 2012. Presented at the 2012 NASA CNS Workshop, January 17, 2012. , Jan-2012
Abstracts for Journals and Proceedings	Wyrobek AJ, Lowe XR, Rabin BM, Marchetti F, Bhatnager S, Northen T, Witkowska E, Straume T, Snijders A. "Molecular Characterization of Long-Term Degenerative CNS Risks from Space Radiation: studies of molecular and cellular defects in the choroid plexus." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Poster presentation. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Articles in Peer-reviewed Journals	Lowe XR, Wyrobek AJ. "Characterization of the early CNS stress biomarkers and profiles associated with neuropsychiatric diseases." Current Genomics. 2012 Sep;13(6):489-97. http://dx.doi.org/10.2174/138920212802510448 ; PubMed PMID: 23449133 , Sep-2012